

VIBEENA R

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Professional Summary

M.Tech Software Engineering student at VIT with experience in machine learning, artificial intelligence, data science, and cloud-based application development. Skilled in c/c++, Python, TensorFlow, Scikit-learn, SQL, AWS, Docker, REST APIs, and CI/CD, with hands-on experience building predictive models, developing end-to-end ML pipelines, and deploying scalable AI solutions. Proficient in data analysis, feature engineering, model evaluation, system integration, and MLOps practices, with a strong focus on delivering reliable, data-driven, and production-ready applications.

Experience

Python Development Intern | RedBack IT Solutions, Vellore

Feb 2026 – Mar 2026

- Built automated **ETL pipelines** (Pandas, NumPy) to process structured datasets; authored architectural comparison reports communicating performance trade-offs to engineering stakeholders.
- Applied OOP design patterns for modular components; wrote **PyTest** unit tests and maintained codebase with **Git**; produced technical design documentation.

Software Engineering Intern | NLC India Limited — General Hospital, Neyveli

Dec 2025 – Jan 2026

- Built a **PHP/Oracle SQL** staff administration portal on **WAMP Server**; enabled role-restricted prescription viewing, PDF report generation/download, and a real-time audit dashboard for compliance.
- Integrated **RBAC** with least-privilege enforcement; normalised relational schemas; produced system architecture documentation for institutional review.

Projects

TriSense Detection using LLMs and BERT | Python, BERT, Transformers, TensorFlow, Scikit-learn, OpenAI API

- Engineered a multi-sense language understanding pipeline using **BERT** and Large Language Models for semantic extraction, generation, and word-sense disambiguation; optimized Transformer configurations to improve classification performance.
- Benchmarked inference latency and model throughput across multiple architectures; deployed predictions through REST APIs with in-memory caching to reduce redundant LLM inference requests and improve response times.

Potato Leaf Disease Classification — Full Stack AI App | Python, TensorFlow, FastAPI, TensorFlow Serving, GCP, React.js

- Built a full-stack AI application by serving a CNN-based potato leaf disease classification model through **FastAPI** and TensorFlow Serving; containerized services with Docker and deployed on **GCP** for scalable inference.
- Designed asynchronous **API endpoints** with input validation, confidence-score responses, and robust error handling; integrated a **React.js** frontend to enable real-time disease prediction and monitoring.

PCOS AI Diagnostic System — 2D Ultrasound to 3D Reconstruction | Python, OpenCV, VTK, CNN, Linux, SHA-256

- Developed a medical imaging pipeline that reconstructed 3D ovarian volumetric heatmaps from 2D ultrasound slices using **voxel stacking, bilateral filtering, and CNN-based preprocessing** techniques for enhanced follicle visualization.
- Integrated **OpenCV-based ROI extraction and automated follicle detection workflow**; implemented **SHA-256** integrity verification to secure medical image data and ensure reliable diagnostic processing.

Cloud AI Anomaly Detection | Python, AWS (EC2, S3), Scikit-learn, Docker, CI/CD, GitHub Actions

- Built and deployed an end-to-end anomaly detection pipeline on **AWS EC2** and **S3**; automated feature engineering, model versioning, drift monitoring, and inference workflows following **MLOps** best practices.
- Containerized services with Docker and automated retraining through GitHub Actions **CI/CD pipelines**; developed **PyTest-based unit and integration tests** with comprehensive architecture and data-flow documentation.

Technical Skills

- Cloud & DevOps:** AWS (EC2, S3), GCP (familiar), Docker, CI/CD (GitHub Actions), microservices, containerisation
- AI/ML & Data:** Scikit-learn, XGBoost, TensorFlow, PyTorch, BERT/Transformers, MLOps, ETL/ELT, Pandas, NumPy, SQL, OpenCV
- Security & Compliance:** RBAC, REST API security, audit-trail design, SHA-256 data integrity, phishing/threat classification; Cisco Cybersecurity certified
- Languages & Backend:** Python (Advanced), Oracle SQL, PHP, FastAPI, MySQL, MongoDB, React.js, PyTest, Git, Jira
- Technical Communication:** Architectural documentation, data-flow diagrams, performance benchmarking, technical report writing

Education

Vellore Institute of Technology (VIT)

Vellore, India

Integrated M.Tech — Software Engineering

2022 – 2027

CGPA: 7.38/10 | Relevant Coursework: DSA, Database Systems, OS, Software Design Patterns, Computer Architecture

Spring Days CBSE School

Tamil Nadu, India

Higher Secondary (77.2%) | Secondary (81.2%)

Certifications

- [AWS Cloud Technical Essentials — Amazon Web Services](#)
- [IBM Data Science Professional Certificate — IBM \(Python, Gen AI, ML, Data Analysis\)](#)
- [Introduction to Cybersecurity — Cisco](#)
- [Software Engineering Intern Certificate — NLC India Limited, General Hospital](#)
- [Python Core Intern Certificate — RedBack IT Solutions](#)